



1. Description

1.1. Project

Project Name	CoprocessorFW
Board Name	custom
Generated with:	STM32CubeMX 6.17.0
Date	07/08/2026

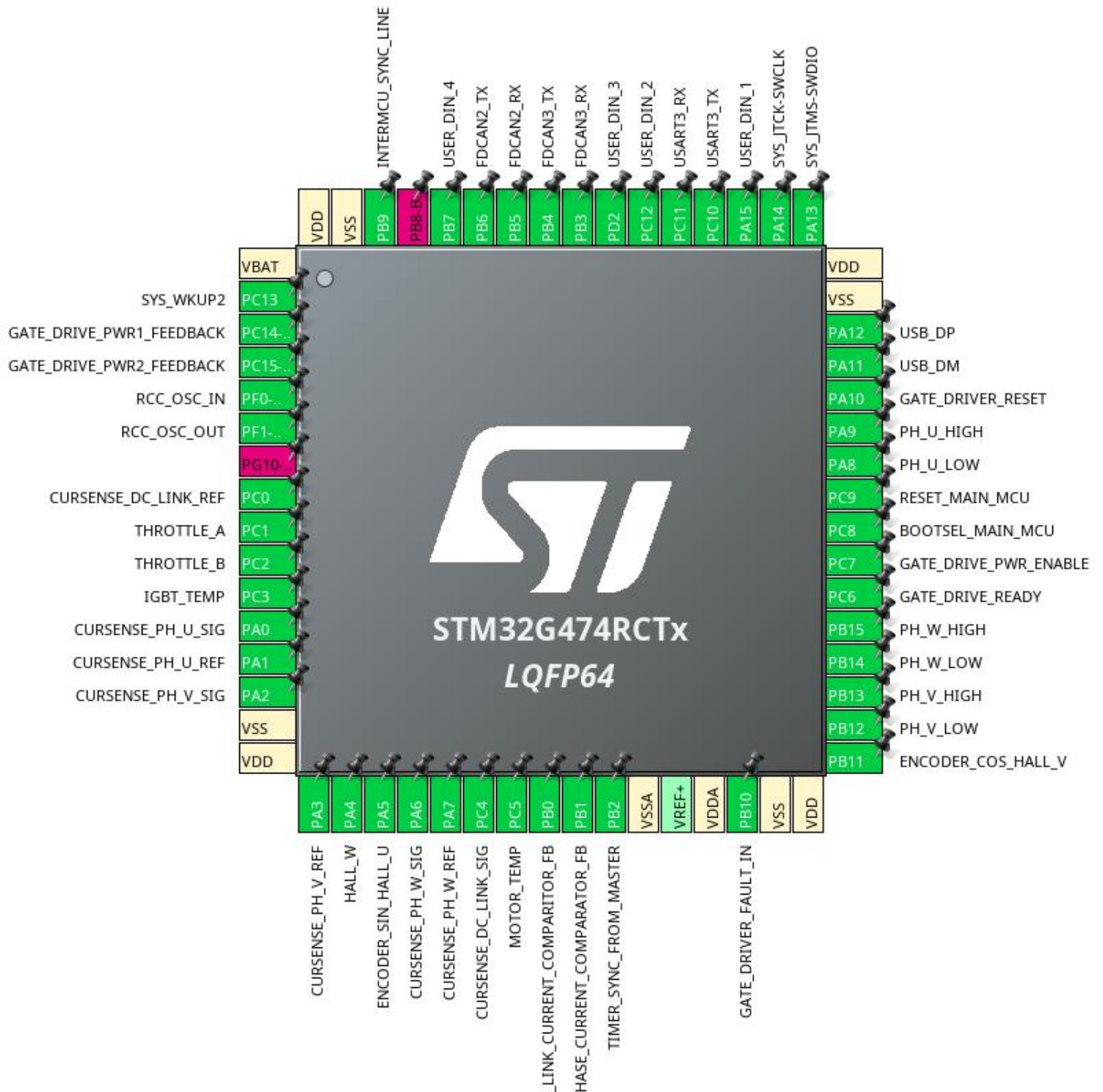
1.2. MCU

MCU Series	STM32G4
MCU Line	STM32G4x4
MCU name	STM32G474RCTx
MCU Package	LQFP64
MCU Pin number	64

1.3. Core(s) information

Core(s)	ARM Cortex-M4
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2. Pinout Configuration



3. Pins Configuration

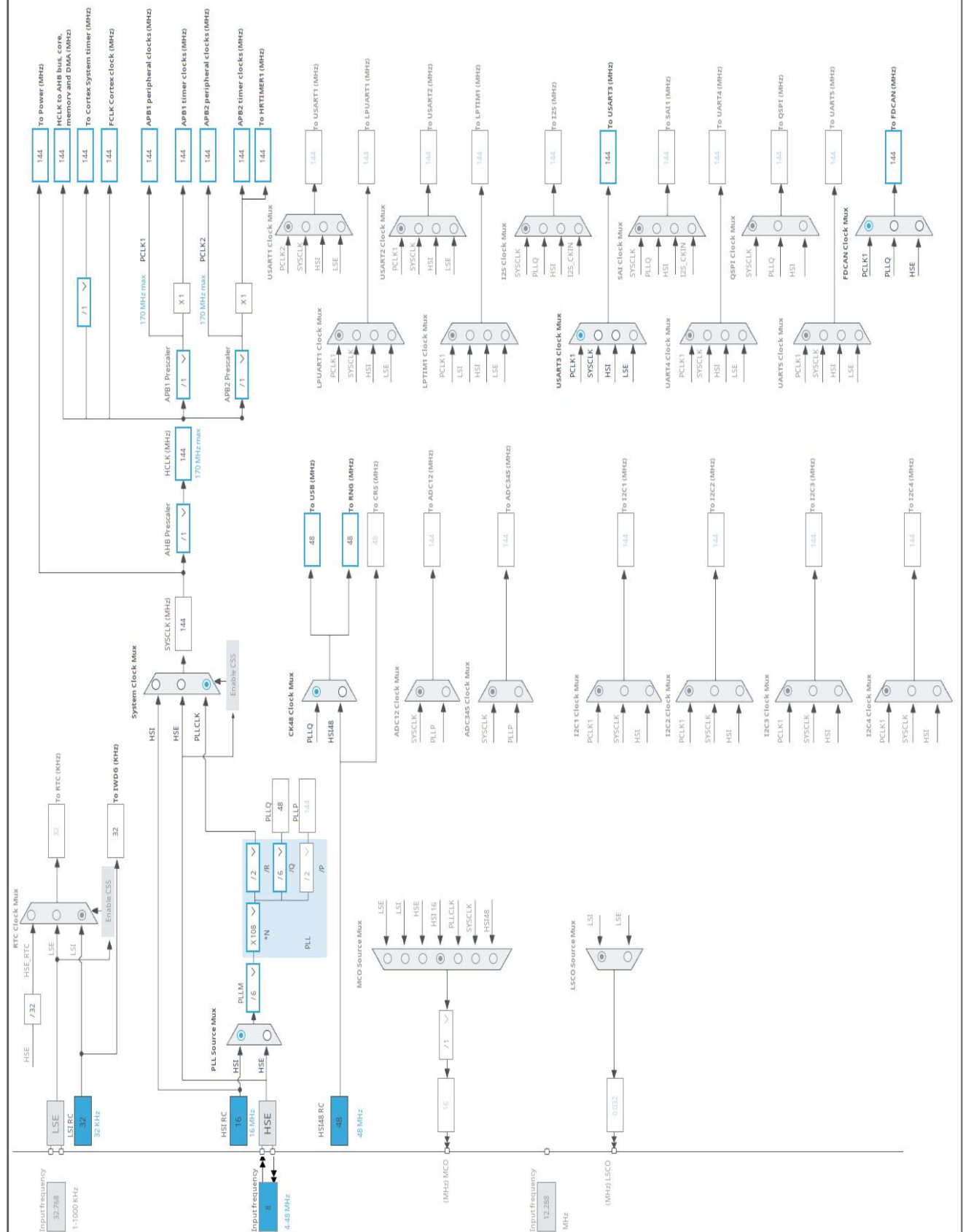
Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
2	PC13	I/O	SYS_WKUP2	
3	PC14-OSC32_IN *	I/O	GPIO_Input	GATE_DRIVE_PWR1_FEE DBACK
4	PC15-OSC32_OUT *	I/O	GPIO_Input	GATE_DRIVE_PWR2_FEE DBACK
5	PF0-OSC_IN	I/O	RCC_OSC_IN	
6	PF1-OSC_OUT	I/O	RCC_OSC_OUT	
7	PG10-NRST	I/O		
8	PC0	I/O	ADC2_IN6	CURSENSE_DC_LINK_RE F
9	PC1	I/O	ADC1_IN7	THROTTLE_A
10	PC2	I/O	ADC1_IN8	THROTTLE_B
11	PC3	I/O	ADC1_IN9	IGBT_TEMP
12	PA0	I/O	ADC1_IN1, COMP3_INP	CURSENSE_PH_U_SIG
13	PA1	I/O	ADC1_IN2	CURSENSE_PH_U_REF
14	PA2	I/O	ADC1_IN3	CURSENSE_PH_V_SIG
15	VSS	Power		
16	VDD	Power		
17	PA3	I/O	ADC1_IN4	CURSENSE_PH_V_REF
18	PA4	I/O	ADC2_IN17	HALL_W
19	PA5	I/O	ADC2_IN13	ENCODER_SIN_HALL_U
20	PA6	I/O	ADC2_IN3	CURSENSE_PH_W_SIG
21	PA7	I/O	ADC2_IN4	CURSENSE_PH_W_REF
22	PC4	I/O	ADC2_IN5	CURSENSE_DC_LINK_SIG
23	PC5	I/O	ADC2_IN11	MOTOR_TEMP
24	PB0	I/O	ADC3_IN12	DC_LINK_CURRENT_COM PARITOR_FB
25	PB1	I/O	GPIO_Analog, ADC3_IN1	PHASE_CURRENT_COMP ARATOR_FB
26	PB2	I/O	HRTIM1_SCIN	TIMER_SYNC_FROM_MAS TER
27	VSSA	Power		
29	VDDA	Power		
30	PB10	I/O	HRTIM1_FLT3	GATE_DRIVER_FAULT_IN
31	VSS	Power		
32	VDD	Power		

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
33	PB11	I/O	ADC2_IN14	ENCODER_COS_HALL_V
34	PB12	I/O	HRTIM1_CHC1	PH_V_LOW
35	PB13	I/O	HRTIM1_CHC2	PH_V_HIGH
36	PB14	I/O	HRTIM1_CHD1	PH_W_LOW
37	PB15	I/O	HRTIM1_CHD2	PH_W_HIGH
38	PC6 *	I/O	GPIO_Input	GATE_DRIVE_READY
39	PC7 *	I/O	GPIO_Output	GATE_DRIVE_PWR_ENABLE
40	PC8 *	I/O	GPIO_Output	BOOTSEL_MAIN_MCU
41	PC9 *	I/O	GPIO_Output	RESET_MAIN_MCU
42	PA8	I/O	HRTIM1_CHA1	PH_U_LOW
43	PA9	I/O	HRTIM1_CHA2	PH_U_HIGH
44	PA10 *	I/O	GPIO_Output	GATE_DRIVER_RESET
45	PA11	I/O	USB_DM	
46	PA12	I/O	USB_DP	
47	VSS	Power		
48	VDD	Power		
49	PA13	I/O	SYS_JTMS-SWDIO	
50	PA14	I/O	SYS_JTCK-SWCLK	
51	PA15 *	I/O	GPIO_Input	USER_DIN_1
52	PC10	I/O	USART3_TX	
53	PC11	I/O	USART3_RX	
54	PC12 *	I/O	GPIO_Input	USER_DIN_2
55	PD2 *	I/O	GPIO_Input	USER_DIN_3
56	PB3	I/O	FDCAN3_RX	
57	PB4	I/O	FDCAN3_TX	
58	PB5	I/O	FDCAN2_RX	
59	PB6	I/O	FDCAN2_TX	
60	PB7 *	I/O	GPIO_Input	USER_DIN_4
61	PB8-BOOT0	I/O		
62	PB9 *	I/O	GPIO_Input	INTERMCU_SYNC_LINE
63	VSS	Power		
64	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration

RIF configuration not available for STM32G474RCTx



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32G4
Line	STM32G4x4
MCU	STM32G474RCTx
Datasheet	DS12288 Rev0

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

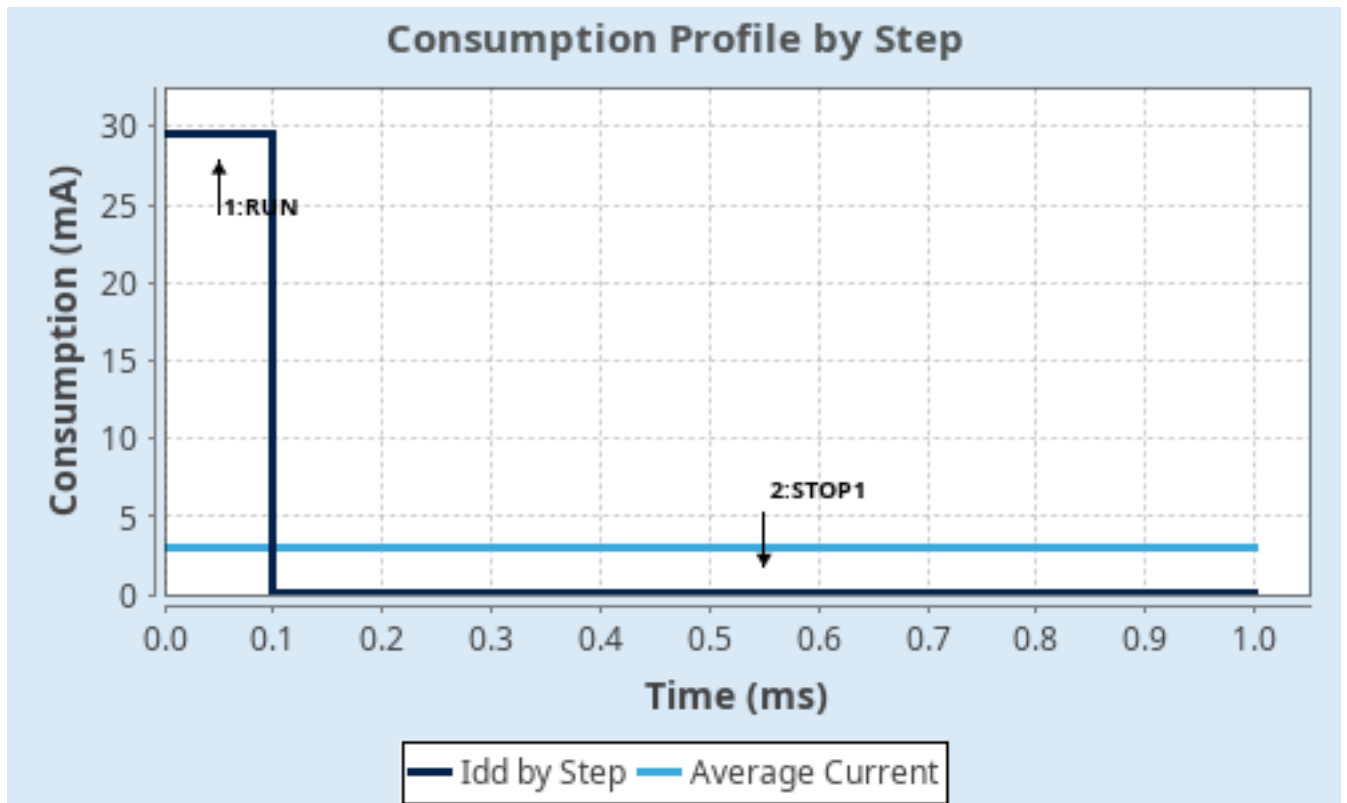
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP1
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	Range1-Boost	NoRange
Fetch Type	FLASH/DualBank/ART	NA
CPU Frequency	170 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL CLOCKS OFF
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	29.5 mA	80.5 μ A
Duration	0.1 ms	0.9 ms
DMIPS	213.0	0.0
Ta Max	124.25	129.98
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	3.02 mA
Battery Life	1 month, 16 days, 9 hours	Average DMIPS	212.5 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	CoprocessorFW
Project Folder	/home/tliao/Desktop/InverterGen5/CoprocessorFW
Toolchain / IDE	EWARM V8.50
Firmware Package Name and Version	STM32Cube FW_G4 V1.6.3
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy all used libraries into the project folder
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_ADC1_Init	ADC1
4	MX_ADC2_Init	ADC2
5	MX_FDCAN2_Init	FDCAN2
6	MX_FDCAN3_Init	FDCAN3
7	MX_HRTIM1_Init	HRTIM1
8	MX_USART3_UART_Init	USART3
9	MX_USB_PCD_Init	USB
10	MX_WWDG_Init	WWDG
11	MX_ADC3_Init	ADC3

Rank	Function Name	Peripheral Instance Name
12	MX_COMP3_Init	COMP3
13	MX_CORDIC_Init	CORDIC
14	MX_CRC_Init	CRC
15	MX_DAC1_Init	DAC1
16	MX_FMAC_Init	FMAC
17	MX_IWDG_Init	IWDG
18	MX_RNG_Init	RNG
19	MX_TIM1_Init	TIM1

3. Peripherals and Middlewares Configuration

3.1. ADC1

IN1: IN1 Single-ended

IN2: IN2 Single-ended

IN3: IN3 Single-ended

IN4: IN4 Single-ended

IN7: IN7 Single-ended

IN8: IN8 Single-ended

IN9: IN9 Single-ended

mode: Temperature Sensor Channel

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode Independent mode

ADC_Settings:

Clock Prescaler Synchronous clock mode divided by 4

Resolution ADC 12-bit resolution

Data Alignment Right alignment

Gain Compensation 0

Scan Conversion Mode Disabled

End Of Conversion Selection End of single conversion

Low Power Auto Wait Disabled

Continuous Conversion Mode Disabled

Discontinuous Conversion Mode Disabled

DMA Continuous Requests Disabled

Overrun behaviour Overrun data preserved

ADC_Regular_ConversionMode:

Enable Regular Conversions Enable

Enable Regular Oversampling Disable

Number Of Conversion 1

External Trigger Conversion Source Regular Conversion launched by software

External Trigger Conversion Edge None

Rank 1

Channel **Channel 2 ***

Sampling Time 2.5 Cycles

Offset Number No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode false

3.2. ADC2

IN3: IN3 Single-ended

IN4: IN4 Single-ended

IN5: IN5 Single-ended

IN6: IN6 Single-ended

IN11: IN11 Single-ended

IN13: IN13 Single-ended

IN14: IN14 Single-ended

mode: IN17 Single-ended

3.2.1. Parameter Settings:

ADCs_Common_Settings:

Mode Independent mode

ADC_Settings:

Clock Prescaler Synchronous clock mode divided by 4

Resolution ADC 12-bit resolution

Data Alignment Right alignment

Gain Compensation 0

Scan Conversion Mode Disabled

End Of Conversion Selection End of single conversion

Low Power Auto Wait Disabled

Continuous Conversion Mode Disabled

Discontinuous Conversion Mode Disabled

DMA Continuous Requests Disabled

Overrun behaviour Overrun data preserved

ADC_Regular_ConversionMode:

Enable Regular Conversions Enable

Enable Regular Oversampling Disable

Number Of Conversion 1

External Trigger Conversion Source Regular Conversion launched by software

External Trigger Conversion Edge None

Rank 1

Channel	Channel 3
Sampling Time	2.5 Cycles
Offset Number	No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable
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Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false
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Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false
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Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false
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3.3. ADC3

mode: IN1

mode: IN12

3.3.1. Parameter Settings:

ADCs_Common_Settings:

Mode	Independent mode
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ADC_Settings:

Clock Prescaler	Synchronous clock mode divided by 4
Resolution	ADC 12-bit resolution
Data Alignment	Right alignment
Gain Compensation	0
Scan Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Low Power Auto Wait	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
Overrun behaviour	Overrun data preserved

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
<u>Rank</u>	1
Channel	Channel 12 *

Sampling Time	2.5 Cycles
Offset Number	No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable
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Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false
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Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false
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Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false
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3.4. COMP3

mode: Input [+]

3.4.1. Parameter Settings:

Basic Parameters:

Trigger Mode	None
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Hysteresis Level	None
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Output Configuration:

Blanking Source	None
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Output Polarity	COMP output on GPIO isn't inverted
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3.5. CORDIC

mode: Activated

3.6. CRC

mode: Activated

3.6.1. Parameter Settings:

Basic Parameters:

Default Polynomial State	Enable
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Default Init Value State	Enable
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Advanced Parameters:

Input Data Inversion Mode	None
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Output Data Inversion Mode	Disable
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Input Data Format	Bytes
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3.7. DAC1

OUT1 mode: OUT1 Connected to on chip-peripherals only

3.7.1. Parameter Settings:

DAC Out1 Settings:

Mode selected	Normal Mode
Output Buffer	Disable
DAC High Frequency	Mode Automatic
DMA Double Data	Disable
Signed Format	Disable
Trigger	None
Trigger2	None
User Trimming	Factory trimming

3.8. FDCAN2

mode: Activated

3.8.1. Parameter Settings:

Basic Parameters:

Clock Divider	Divide kernel clock by 1
Frame Format	Classic mode
Mode	Normal mode
Auto Retransmission	Disable
Transmit Pause	Disable
Protocol Exception	Disable
Nominal Sync Jump Width	1
Data Prescaler	1
Data Sync Jump Width	1
Data Time Seg1	1
Data Time Seg2	1
Std Filters Nbr	0
Ext Filters Nbr	0
Tx Fifo Queue Mode	FIFO mode

Bit Timings Parameters:

Nominal Prescaler	16
Nominal Time Quantum	111.11111111111111 *
Nominal Time Seg1	1

Nominal Time Seg2	1
Nominal Time for one Bit	333 *
Nominal Baud Rate	2999999 *

3.9. FDCAN3

mode: Activated

3.9.1. Parameter Settings:

Basic Parameters:

Clock Divider	Divide kernel clock by 1
Frame Format	Classic mode
Mode	Normal mode
Auto Retransmission	Disable
Transmit Pause	Disable
Protocol Exception	Disable
Nominal Sync Jump Width	1
Data Prescaler	1
Data Sync Jump Width	1
Data Time Seg1	1
Data Time Seg2	1
Std Filters Nbr	0
Ext Filters Nbr	0
Tx Fifo Queue Mode	FIFO mode

Bit Timings Parameters:

Nominal Prescaler	16
Nominal Time Quantum	111.11111111111111 *
Nominal Time Seg1	1
Nominal Time Seg2	1
Nominal Time for one Bit	333 *
Nominal Baud Rate	2999999 *

3.10. FMAC

mode: Mode

3.11. HRTIM1

Timer A: TA1 and TA2 outputs active

Timer C: TC1 and TC2 outputs active

Timer D: TD1 and TD2 outputs active

mode: Fault Input 3

External Synchronization: Enable Input Synchronization PIN

3.11.1. HRTIM Interrupt Configuration:

Sources:

1st Source of interrupt	No interrupt enabled
2nd Source of interrupt	No interrupt enabled
3rd Source of interrupt	No interrupt enabled
4th Source of interrupt	No interrupt enabled
5th Source of interrupt	No interrupt enabled
6th Source of interrupt	No interrupt enabled
7th Source of interrupt	No interrupt enabled
8th Source of interrupt	No interrupt enabled
9th Source of interrupt	No interrupt enabled

3.11.2. Synchro Configuration:

Master Timer Synchronization:

Sync Options	HRTIM instance doesn't handle external synchronization signals (SYNCIN, SYNCOUT)
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3.11.3. High Resolution:

High Resolution Setting:

High Resolution to reach steps down to 184 ps (5.44GHz Equivalent Frequency for HRTIMclock=170MHz)	Enabled with Simple DLL Calibration
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Calibration Settings:

Calibration Rate	Periodic DLL calibration: T : 2048 mult tHRTIM (0.012 ms)
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Polling:

Timeout (in ms)	10
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3.11.4. External Event Configuration:

External Event 1:

Event Configuration	Disable
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External Event 2:

Event Configuration Disable

External Event 3:

Event Configuration Disable

External Event 4:

Event Configuration Disable

External Event 5:

Event Configuration Disable

External Event 6:

Event Configuration Disable

External Event 7:

Event Configuration Disable

External Event 8:

Event Configuration Disable

External Event 9:

Event Configuration Disable

External Event 10:

Event Configuration Disable

3.11.5. Fault Lines Configuration:

Common Prescaler:

Prescaler fFLTS=fHRTIM

Fault Line 1:

Line Configuration No Configuration of Fault Line

Fault Line 2:

Line Configuration No Configuration of Fault Line

Fault Line 3:

Line Configuration Configure Fault Line

Source Fault input is FLT input pin

Polarity External event is active high

Filter Filter disabled

Lock Fault settings bits are read/write

Line Ctl Enable Fault Line

Counter on Faults (Threshold) 0

Reset Mode Fault counter is reset on each reset / roll-over event

Blanking Source Type Reset-Aligned Window - Blanking window starts on Timer C reset/roll-over and ends on Timer C COMP 3 event

Fault Line 4:

Line Configuration No Configuration of Fault Line

Fault Line 5:

Line Configuration	No Configuration of Fault Line
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Fault Line 6:

Line Configuration	No Configuration of Fault Line
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3.11.6. ADC Triggers Configuration:

ADC Trigger 1:

ADC Trigger Configuration	Disable
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ADC Trigger 2:

ADC Trigger Configuration	Disable
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ADC Trigger 3:

ADC Trigger Configuration	Disable
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ADC Trigger 4:

ADC Trigger Configuration	Disable
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ADC Trigger 5:

ADC Trigger Configuration	Disable
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ADC Trigger 6:

ADC Trigger Configuration	Disable
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ADC Trigger 7:

ADC Trigger Configuration	Disable
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ADC Trigger 8:

ADC Trigger Configuration	Disable
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ADC Trigger 9:

ADC Trigger Configuration	Disable
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ADC Trigger 10:

ADC Trigger Configuration	Disable
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3.11.7. Burst Mode Configuration:

Burst Mode Enabling:

Burst Mode	Burst mode disabled
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3.11.8. Timer A:

General:

Timer Idx	Timer A
Basic/Advanced Configuration	Advanced (using HAL_Waveform methods)

Time Base Setting:

Prescaler Ratio	HRTIM Clock Multiplied by 32 (HRTIM Clock is set in Clock Configuration Tab with Max Value = 170MHz)
fHRCK Equivalent Frequency	4.608E9
Period	0xFFDF *
Resulting PWM Frequency	70348
Repetition Counter	0x00 *
Up Down Mode	Timer counter is operating in up-counting mode
Mode	The timer operates in continuous (free-running) mode
Timing Unit:	
Interleaved Mode	Disabled
Start On Sync	Synchronization input event has no effect on the timer
Reset On Sync	Synchronization input event has no effect on the timer
Dac Synchro	No DAC synchronization event generated
Preload Enable	Preload disabled: the write access is directly done into the active register
Update Gating	Update done independently from the DMA burst transfer completion
Repetition Update	Update on repetition disabled
Burst Mode	Timer counter clock is maintained and the timer operates normally
Push Pull	Push-Pull mode disabled
Number of Faults to enable	0
Fault Lock	Timer fault enabling bits are read/write
Dead Time Insertion	Output 1 and output 2 signals are independent
Delayed Protection Mode	No action
Update Trigger Sources Selection : Please enter the number of Triggers to select	0
Reset Update	Update by Timer reset / roll-over disabled
Resynchronized Update	Update taken into account immediately
Reset Trigger Sources Selection : Please enter the number of Triggers to select	0
Interrupt Requests Sources Selection : Please enter the number of Active Interrupt Requests	0
Number of Timer A Internal DMA Request Sources - you first have to enable the Timer A DMA Request in the DMA Settings Tab	0
Compare Unit 1:	
Compare Unit 1 Configuration	Disable
Compare Unit 2:	
Compare Unit 2 Configuration	Disable
Compare Unit 3:	
Compare Unit 3 Configuration	Disable
Compare Unit 4:	
Compare Unit 4 Configuration	Disable
Capture Unit 1:	
Capture Unit 1 Configuration	Disable

Capture Unit 2:

Capture Unit 2 Configuration Disable

External Event 1 Filtering:

Filtering Configuration Disable

External Event 2 Filtering:

Filtering Configuration Disable

External Event 3 Filtering:

Filtering Configuration Disable

External Event 4 Filtering:

Filtering Configuration Disable

External Event 5 Filtering:

Filtering Configuration Disable

External Event 6 Filtering:

Filtering Configuration Disable

External Event 7 Filtering:

Filtering Configuration Disable

External Event 8 Filtering:

Filtering Configuration Disable

External Event 9 Filtering:

Filtering Configuration Disable

External Event 10 Filtering:

Filtering Configuration Disable

External Event Counter A:

Event Counter A Configuration Disable

Burst DMA Controller:

Burst DMA Configuration Disable

Dead Time:

Dead Time Configuration Disable

Swap Output1 and Output2:

TA1 Output is sensitive to TA2 Control Registers and
vice versa Disable

Output 1 Configuration:

Output1 Configuration TA1

Polarity Output is active HIGH

Set Source Selection : Please enter the number of
Active Set Sources 0

Reset Source Selection : Please enter the number of
Active Reset Sources 0

Idle Mode The output is not affected by the burst mode operation

Idle Level Output at inactive level when in IDLE state

Fault Level	The output is not affected by the fault input
Chopper Mode Enable	Output signal is not altered
Burst Mode Entry Delayed	The programmed Idle state is applied immediately to the Output

Output 2 Configuration:

Output2 Configuration	TA2
Polarity	Output is active HIGH
Set Sources: nothing to set as Dead Time is enabled,	0
Output2 signal is defined based on Output1	
Configuration Set Sources	
Reset Source Selection : Please enter the number of	0
Active Reset Sources	
Idle Mode	The output is not affected by the burst mode operation
Idle Level	Output at inactive level when in IDLE state
Fault Level	The output is not affected by the fault input
Chopper Mode Enable	Output signal is not altered
Burst Mode Entry Delayed	The programmed Idle state is applied immediately to the Output

Chopper Mode:

Chopper Mode Configuration	Disable
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Dual DAC Channel Configuration:

Dual DAC Channel Trigger	Disabled
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3.11.9. Timer C:

General:

Timer Idx	Timer C
Basic/Advanced Configuration	Advanced (using HAL_Waveform methods)

Time Base Setting:

Prescaler Ratio	HRTIM Clock Multiplied by 32 (HRTIM Clock is set in Clock Configuration Tab with Max Value = 170MHz)
fHRCK Equivalent Frequency	4.608E9
Period	0xFFDF *
Resulting PWM Frequency	70348
Repetition Counter	0x00 *
Up Down Mode	Timer counter is operating in up-counting mode
Mode	The timer operates in continuous (free-running) mode

Timing Unit:

Interleaved Mode	Disabled
Start On Sync	Synchronization input event has no effect on the timer
Reset On Sync	Synchronization input event has no effect on the timer
Dac Synchro	No DAC synchronization event generated
Preload Enable	Preload disabled: the write access is directly done into the active register
Update Gating	Update done independently from the DMA burst transfer completion

Repetition Update	Update on repetition disabled
Burst Mode	Timer counter clock is maintained and the timer operates normally
Push Pull	Push-Pull mode disabled
Number of Faults to enable	0
Fault Lock	Timer fault enabling bits are read/write
Dead Time Insertion	Output 1 and output 2 signals are independent
Delayed Protection Mode	No action
Update Trigger Sources Selection : Please enter the number of Triggers to select	0
Reset Update	Update by Timer reset / roll-over disabled
Resynchronized Update	Update taken into account immediately
Reset Trigger Sources Selection : Please enter the number of Triggers to select	0
Interrupt Requests Sources Selection : Please enter the number of Active Interrupt Requests	0
Number of Timer C Internal DMA Request Sources - you first have to enable the Timer C DMA Request in the DMA Settings Tab	0
Compare Unit 1:	
Compare Unit 1 Configuration	Disable
Compare Unit 2:	
Compare Unit 2 Configuration	Disable
Compare Unit 3:	
Compare Unit 3 Configuration	Disable
Compare Unit 4:	
Compare Unit 4 Configuration	Disable
Capture Unit 1:	
Capture Unit 1 Configuration	Disable
Capture Unit 2:	
Capture Unit 2 Configuration	Disable
External Event 1 Filtering:	
Filtering Configuration	Disable
External Event 2 Filtering:	
Filtering Configuration	Disable
External Event 3 Filtering:	
Filtering Configuration	Disable
External Event 4 Filtering:	
Filtering Configuration	Disable
External Event 5 Filtering:	
Filtering Configuration	Disable
External Event 6 Filtering:	
Filtering Configuration	Disable

Chopper Mode:

Chopper Mode Configuration Disable

Dual DAC Channel Configuration:

Dual DAC Channel Trigger Disabled

3.11.10. Timer D:

General:

Timer Idx Timer D
Basic/Advanced Configuration Advanced (using HAL_Waveform methods)

Time Base Setting:

Prescaler Ratio HRTIM Clock Multiplied by 32 (HRTIM Clock is set in Clock Configuration Tab with Max Value = 170MHz)

 fHRCK Equivalent Frequency 4.608E9

Period **0xFFDF ***

 Resulting PWM Frequency 70348

Repetition Counter **0x00 ***

Up Down Mode Timer counter is operating in up-counting mode

Mode The timer operates in continuous (free-running) mode

Timing Unit:

Interleaved Mode Disabled

Start On Sync Synchronization input event has no effect on the timer

Reset On Sync Synchronization input event has no effect on the timer

Dac Synchro No DAC synchronization event generated

Preload Enable Preload disabled: the write access is directly done into the active register

Update Gating Update done independently from the DMA burst transfer completion

Repetition Update Update on repetition disabled

Burst Mode Timer counter clock is maintained and the timer operates normally

Push Pull Push-Pull mode disabled

Number of Faults to enable 0

Fault Lock Timer fault enabling bits are read/write

Dead Time Insertion Output 1 and output 2 signals are independent

Delayed Protection Mode No action

Update Trigger Sources Selection : Please enter the number of Triggers to select 0

Reset Update Update by Timer reset / roll-over disabled

Resynchronized Update Update taken into account immediately

Reset Trigger Sources Selection : Please enter the number of Triggers to select 0

Interrupt Requests Sources Selection : Please enter the number of Active Interrupt Requests 0

Number of Timer D Internal DMA Request Sources - 0

you first have to enable the Timer D DMA Request in the DMA Settings Tab

Compare Unit 1:

Compare Unit 1 Configuration	Disable
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Compare Unit 2:

Compare Unit 2 Configuration	Disable
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Compare Unit 3:

Compare Unit 3 Configuration	Disable
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Compare Unit 4:

Compare Unit 4 Configuration	Disable
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Capture Unit 1:

Capture Unit 1 Configuration	Disable
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Capture Unit 2:

Capture Unit 2 Configuration	Disable
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External Event 1 Filtering:

Filtering Configuration	Disable
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External Event 2 Filtering:

Filtering Configuration	Disable
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External Event 3 Filtering:

Filtering Configuration	Disable
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External Event 4 Filtering:

Filtering Configuration	Disable
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External Event 5 Filtering:

Filtering Configuration	Disable
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External Event 6 Filtering:

Filtering Configuration	Disable
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External Event 7 Filtering:

Filtering Configuration	Disable
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External Event 8 Filtering:

Filtering Configuration	Disable
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External Event 9 Filtering:

Filtering Configuration	Disable
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External Event 10 Filtering:

Filtering Configuration	Disable
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External Event Counter A:

Event Counter A Configuration	Disable
-------------------------------	---------

Burst DMA Controller:

Burst DMA Configuration	Disable
-------------------------	---------

Dead Time:

TD1 Output is sensitive to TD2 Control Registers and Disable
vice versa

Output1 Configuration	TD1
Polarity	Output is active HIGH
Set Source Selection : Please enter the number of Active Set Sources	0
Reset Source Selection : Please enter the number of Active Reset Sources	0

Idle Mode	The output is not affected by the burst mode operation
Idle Level	Output at inactive level when in IDLE state
Fault Level	The output is not affected by the fault input
Chopper Mode Enable	Output signal is not altered
Burst Mode Entry Delayed	The programmed Idle state is applied immediately to the Output

Output2 Configuration	TD2
Polarity	Output is active HIGH
Set Sources: nothing to set as Dead Time is enabled,	0
Output2 signal is defined based on Output1	
Configuration Set Sources	
Reset Source Selection : Please enter the number of	0
Active Reset Sources	

Idle Mode	The output is not affected by the burst mode operation
Idle Level	Output at inactive level when in IDLE state
Fault Level	The output is not affected by the fault input
Chopper Mode Enable	Output signal is not altered
Burst Mode Entry Delayed	The programmed Idle state is applied immediately to the Output

Chopper Mode Configuration	Disable
----------------------------	---------

Dual DAC Channel Trigger Disabled

3.12.1. Parameter Settings:

IWDG counter clock prescaler	4
IWDG window value	4095

IWDG down-counter reload value 4095

3.13. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.13.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Disabled
Data Cache	Enabled
Flash Latency(WS)	4 WS (5 CPU cycle)

RCC Parameters:

HSI Calibration Value	64
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 1
-------------------------------	---------------------------------

Peripherals Clock Configuration:

Generate the peripherals clock configuration	TRUE
--	------

3.14. RNG

mode: Activated

3.14.1. Parameter Settings:

Clock Error Detection	Enable
-----------------------	--------

3.15. SYS

Debug: Serial Wire

mode: System Wake-Up 2

Timebase Source: SysTick

mode: save power of non-active UCPD - deactive Dead Battery pull-up

3.16. TIM1

Slave Mode: Trigger Mode

Trigger Source: ITR0

3.16.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable
Slave Mode Controller	Trigger Mode

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

3.17. USART3

Mode: Multiprocessor Communication

3.17.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
Wake-Up Method	Idle Line
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable

RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.18. USB

mode: Device (FS)

3.18.1. Parameter Settings:

Basic Parameters:

Speed	Full Speed 12MBit/s
Physical interface	Internal Phy
Sof Enable	Disabled

Power Parameters:

Low Power	Disabled
Link Power Management	Disabled
Battery Charging	Disabled

3.19. WWDG

mode: Activated

3.19.1. Parameter Settings:

Watchdog Clocking:

WWDG counter clock prescaler	1
WWDG window value	64
WWDG free-running downcounter value	64

Watchdog Interrupt:

Early wakeup interrupt	Disable
------------------------	---------

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PC1	ADC1_IN7	Analog mode	No pull-up and no pull-down	n/a	THROTTLE_A
	PC2	ADC1_IN8	Analog mode	No pull-up and no pull-down	n/a	THROTTLE_B
	PC3	ADC1_IN9	Analog mode	No pull-up and no pull-down	n/a	IGBT_TEMP
	PA0	ADC1_IN1	Analog mode	No pull-up and no pull-down	n/a	CURSENSE_PH_U_SIG
	PA1	ADC1_IN2	Analog mode	No pull-up and no pull-down	n/a	CURSENSE_PH_U_REF
	PA2	ADC1_IN3	Analog mode	No pull-up and no pull-down	n/a	CURSENSE_PH_V_SIG
	PA3	ADC1_IN4	Analog mode	No pull-up and no pull-down	n/a	CURSENSE_PH_V_REF
ADC2	PC0	ADC2_IN6	Analog mode	No pull-up and no pull-down	n/a	CURSENSE_DC_LINK_REF
	PA4	ADC2_IN17	Analog mode	No pull-up and no pull-down	n/a	HALL_W
	PA5	ADC2_IN13	Analog mode	No pull-up and no pull-down	n/a	ENCODER_SIN_HALL_U
	PA6	ADC2_IN3	Analog mode	No pull-up and no pull-down	n/a	CURSENSE_PH_W_SIG
	PA7	ADC2_IN4	Analog mode	No pull-up and no pull-down	n/a	CURSENSE_PH_W_REF
	PC4	ADC2_IN5	Analog mode	No pull-up and no pull-down	n/a	CURSENSE_DC_LINK_SIG
	PC5	ADC2_IN11	Analog mode	No pull-up and no pull-down	n/a	MOTOR_TEMP
ADC3	PB11	ADC2_IN14	Analog mode	No pull-up and no pull-down	n/a	ENCODER_COS_HALL_V
	PB0	ADC3_IN12	Analog mode	No pull-up and no pull-down	n/a	DC_LINK_CURRENT_COMPARATOR_FB
ADC3	PB1	ADC3_IN1	Analog mode	No pull-up and no pull-down	n/a	PHASE_CURRENT_COMPARATOR_FB
COMP3	PA0	COMP3_INP	Analog mode	No pull-up and no pull-down	n/a	CURSENSE_PH_U_SIG
FDCAN2	PB5	FDCAN2_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB6	FDCAN2_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
FDCAN3	PB3	FDCAN3_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB4	FDCAN3_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
HRTIM1	PB2	HRTIM1_SCIN	Alternate Function Push Pull	No pull-up and no pull-down	Low	TIMER_SYNC_FROM_MASTER
	PB10	HRTIM1_FLT3	Alternate Function Push Pull	No pull-up and no pull-down	Low	GATE_DRIVER_FAULT_IN
	PB12	HRTIM1_CHC1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PH_V_LOW
	PB13	HRTIM1_CHC2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PH_V_HIGH
	PB14	HRTIM1_CHD1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PH_W_LOW
	PB15	HRTIM1_CHD2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PH_W_HIGH
	PA8	HRTIM1_CHA1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PH_U_LOW
	PA9	HRTIM1_CHA2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PH_U_HIGH
HRTIM1	PB2	HRTIM1_SCIN	Alternate Function Push Pull	No pull-up and no pull-down	Low	TIMER_SYNC_FROM_MASTER

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PB10	HRTIM1_FLT3	Alternate Function Push Pull	No pull-up and no pull-down	Low	GATE_DRIVER_FAULT_I N
	PB12	HRTIM1_CHC1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PH_V_LOW
	PB13	HRTIM1_CHC2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PH_V_HIGH
	PB14	HRTIM1_CHD1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PH_W_LOW
	PB15	HRTIM1_CHD2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PH_W_HIGH
	PA8	HRTIM1_CHA1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PH_U_LOW
	PA9	HRTIM1_CHA2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	PH_U_HIGH
RCC	PF0-OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
	PF1-OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	
SYS	PC13	SYS_WKUP2	n/a	n/a	n/a	
	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	
USART3	PC10	USART3_TX	Alternate Function Open Drain	No pull-up and no pull-down	Low	
	PC11	USART3_RX	Alternate Function Open Drain	No pull-up and no pull-down	Low	
USB	PA11	USB_DM	n/a	n/a	n/a	
	PA12	USB_DP	n/a	n/a	n/a	
GPIO	PC14-OSC32_IN	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GATE_DRIVE_PWR1_FE EDBACK
	PC15-OSC32_OUT	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GATE_DRIVE_PWR2_FE EDBACK
	PB1	GPIO_Analog	Analog mode	No pull-up and no pull-down	n/a	PHASE_CURRENT_COM PARATOR_FB
	PC6	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GATE_DRIVE_READY
	PC7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GATE_DRIVE_PWR_ENA BLE
	PC8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	BOOTSEL_MAIN_MCU
	PC9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RESET_MAIN_MCU
	PA10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GATE_DRIVER_RESET
	PA15	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	USER_DIN_1
	PC12	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	USER_DIN_2
	PD2	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	USER_DIN_3
	PB7	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	USER_DIN_4
	PB9	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	INTERMCU_SYNC_LINE

4.2. DMA configuration

nothing configured in DMA service

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
Window watchdog interrupt	unused		
PVD/PVM1/PVM2/PVM3/PVM4 interrupts through EXTI lines 16/38/39/40/41	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
ADC1 and ADC2 global interrupt	unused		
USB high priority interrupt remap	unused		
USB low priority interrupt remap	unused		
TIM1 break interrupt and TIM15 global interrupt	unused		
TIM1 update interrupt and TIM16 global interrupt	unused		
TIM1 trigger and commutation interrupts and TIM17 global interrupt	unused		
TIM1 capture compare interrupt	unused		
USART3 global interrupt / USART3 wake-up interrupt through EXTI line 28	unused		
ADC3 global interrupt	unused		
TIM6 global interrupt, DAC1 and DAC3 channel underrun error interrupts	unused		
COMP1, COMP2 and COMP3 interrupts through EXTI lines 21, 22 and 29	unused		
HRTIM master timer global interrupt	unused		
HRTIM timer A global interrupt	unused		
HRTIM timer C global interrupt	unused		
HRTIM timer D global interrupt	unused		
HRTIM fault global interrupt	unused		
FPU global interrupt	unused		
FDCAN2 interrupt 0	unused		
FDCAN2 interrupt 1	unused		
FDCAN3 interrupt 0	unused		

Interrupt Table	Enable	Preenmption Priority	SubPriority
FDCAN3 interrupt 1		unused	
RNG global interrupt		unused	
CORDIC interrupt		unused	
FMAC interrupt		unused	

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true

* User modified value

5. System Views

5.1. Category view

5.1.1. Current

Middleware							
System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Utilities
DMA	ADC1 ✓	HRTIM1 ✓	FDCAN2 ✓		RNG ✓	CORDIC ✓	
GPIO ✓	ADC2 ✓	TIM1 ✓	FDCAN3 ✓			CRC ✓	
IWDG ✓	ADC3 ✓		USART3 ✓			FMAC ✓	
NVIC ✓	COMP3 ✗		USB ✓				
RCC ✓	DAC1 ✓						
SYS ✓							
WWDG ✓							

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32g4_bsd1.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32g4_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32g4_svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-usb-c-pd-solutions-presentation.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32g4-series-product-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/humanoid-robot-reference-guide.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32g4.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flpowerstbd.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32c5.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-

guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4232-getting-started-with-analog-comparators-for-stm32f3-series-and-stm32g4-series-devices-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4296-use-stm32f3stm32g4-ccm-sram-with-iar-embedded-workbench-keil-mdkarm-stmicroelectronics-stm32cubeide-and-other-gnubased-toolchains-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4539-hrtim-cookbook-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an5315-stm32cube-firmware-examples-for-stm32g4-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5346-stm32g4-adc-

use-tips-and-recommendations-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5788-stm32-digital-power-pid-and-iir-filters-for-smmps-control-design-and-comparison-on-bg414edpow1-discovery-kit-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5325-how-to-use-the-cordic-to-perform-mathematical-functions-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4879-introduction-to-usb-hardware-and-pcb-guidelines-using-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4894-how-to-use-eeeprom-emulation-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-oversampling-techniques-to-improve-signal-to-noise-ratio-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5405-how-to-use-fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5978-introduction-to-mb1971-llc-hat-12-v-to-75-v1-a-for-f334-g474-nucleo-board-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an2548-introduction-to-dma-controller-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4277-how-to-use-pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4635-how-to-optimize-lpuart-power-consumption-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4908-getting-started-with-usart-automatic-baud-rater-detection-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5156-introduction-to-security-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5224-introduction-to-dmamux-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5543-guidelines-for-enhanced-spi-communication-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5348-introduction-to-fdcan-peripherals-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/cd00211314-how-to-optimize-the-adc-accuracy-in-the-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an2639-soldering-recommendations-and-package-information-for-leadfree-ecopack2-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4566-how-to-extend-

the-dac-performance-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5647-the-goertzel-algorithm-to-compute-individual-terms-of-the-discrete-fourier-transform-dft-in-stm32-products-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an2606-introduction-to-system-memory-boot-mode-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3126-how-to-use-dac-for-audio-and-waveform-generation-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3155-how-to-use-usart-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3156-how-to-use-usb-dfu-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4286-how-to-use-spi-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4760-introduction-to-quadspi-interface-for-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4776-how-to-use-generalpurpose-timer-peripheral-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4989-introduction-to-debug-toolbox-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5094-migrating-between-stm32f334303-and-stm32g431g474g491-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5738-guidelines-for-lifetime-estimation-on-stm32g4-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an6363-introduction-to-clock-requirements-and-calibration-for-stm32-mcus-stmicroelectronics.pdf

Errata Sheets	https://www.st.com/resource/en/errata_sheet/es0430-stm32g471xx473xx474xx483xx484xx-device-errata-stmicroelectronics.pdf
Datasheet	https://www.st.com/resource/en/datasheet/dm00431551.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0214-stm32-cortexm4-mcus-and-mpus-programming-manual-stmicroelectronics.pdf
Reference Manuals	https://www.st.com/resource/en/reference_manual/rm0440-stm32g4-series-advanced-armbased-32bit-mcus-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1208-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-tssop-and-ssop-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf
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User Manuals	https://www.st.com/resource/en/user_manual/um3167-stm32g4-series-ulcsaiec-607301603351-selftest-library-user-guide-stmicroelectronics.pdf